

Algebra 1
Unit 8
Lesson 3 Practice

Name _____
Date _____
Period _____

Complete the table for each equation in 1-5.

Equation	Form	Key Feature(s) Revealed	Value(s) of Key Feature
1. $y = (x - 9)(x + 12)$	☐ Vertex form ☐ Factored form ☐ Standard form		
2. $y = (x - 5)^2 - 8$	☐ Vertex form ☐ Factored form ☐ Standard form		
3. $y = -\frac{1}{4}x^2 - 4x - 4$	☐ Vertex form ☐ Factored form ☐ Standard form		
4. $y = 7(x - 1)^2$	☐ Vertex form ☐ Factored form ☐ Standard form		
5. $y = -x^2 + 12x + 36$	☐ Vertex form ☐ Factored form ☐ Standard form		

6. Rewrite the quadratic equation in factored form.

$$y = x^2 - 3x - 10$$

7. Rewrite the quadratic equation in standard form.

$$y = (x - 3)^2 - 7$$

8. Rewrite the equation in factored form.

$$y = -x^2 + 11x - 28$$

9. Rewrite the equation in standard form.

$$y = -2(x - 1)(x + 9)$$

10. Cyrus makes model rockets with his boy scout troop. The rockets launch straight up from 25 meters above the ground with a velocity of 20 meters/second. The height of the rocket is represented by the equation $y = -5x^2 + 20x + 25$, where x is time in seconds. Rewrite the equation in factored form.